HEARING AID USING PCB

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TABLE OF CONTENTS

ABSTRACT:

The objective of this project is to create a printed circuit board (PCB)-based hearing aid for those with hearing loss. The system creates an accessible and economical solution by combining easily available components with unique hardware design. Modules for communication, amplification, and signal processing are all integrated together centrally on the PCB. The project places a strong emphasis on user customisation, enabling users to fit the device exactly to meet their unique hearing requirements. This hearing aid aims to provide users the ability to actively manage their hearing health by utilizing open-source technologies and intuitive interfaces. The project's potential influence on solving the global issue of hearing loss is highlighted in the abstract, especially in areas where access to commercial hearing aids is restricted.

INTRODUCTION

The Hearing Aid Project provides a cutting-edge, easily accessible answer to a pressing need for those with hearing impairments. The goal of this project is to create a cutting-edge hearing aid that is both reasonably priced and equipped with cutting-edge technology. By utilizing developments in acoustics, microelectronics, and signal processing, the project aims to improve user aural experience. The goal is to design a gadget that has functions like noise cancellation and adaptive adjustments in addition to sound amplification. The project's dedication to inclusivity is emphasized in the introduction, guaranteeing that this hearing aid is not only practical but also affordable for a wide range of socioeconomic backgrounds.

A tiny gadget inserted into or around the ear to help people with hearing loss hear better is called an electronic hearing aid. A microphone, signal conditioning, a receiver—also referred to as a speaker—and a battery are the essential parts of a hearing aid. Sound is transformed into an electric signal by the microphone. The signal is then conditioned, which can range from a basic equalization that involves amplifying every sound source equally to a more complex equalization that uses a digital signal processor. The electronics are powered by the battery and the receiver converts the electronic signal back to sound.

Technology Types

Analog and digital technology are the two main categories of hearing aid technology. Analog hearing aids were the first to process sound waves in the analog domain; digital hearing aids, which have been around for a while, process sound waves in the digital domain. The first analog hearing aids were made by amplifying sound and speech, and were prescribed after a patient had undergone testing to identify the specific frequency response that the patient required. Modern analog hearing aids come with the option to be programmed during the fitting process, and some even have several listening profiles that the patient can choose from by pressing a button. In addition, patients can choose from a variety of listening profiles on digital hearing aids, which can be programmed during the fitting process. Sound digitization makes more sophisticated signal processing possible.Sound digitization makes it possible to perform more sophisticated signal processing techniques like filtering, noise reduction, and control over acoustic feedback (ringing). Due to their superior performance and versatility over analog models, the great majority of hearing aids on the market today are digital.

With the goal of revolutionizing hearing aids, the Hearing Aid on PCB Board project is a creative combination of audiology and cutting edge electronics. The creation of an advanced hearing aid that is integrated directly onto a printed circuit board (PCB) is the main goal of this project. The objective is to develop a small, effective, and affordable solution for people with hearing loss by utilizing the design flexibility of PCBs.

In order to maximize the functionality of the hearing aid, the project takes a multidisciplinary approach, exploring acoustics, microelectronics design, and signal processing algorithms. The complex circuitry that improves sound quality, lower background noise, and offers programmable settings to suit different hearing profiles is housed on the PCB.

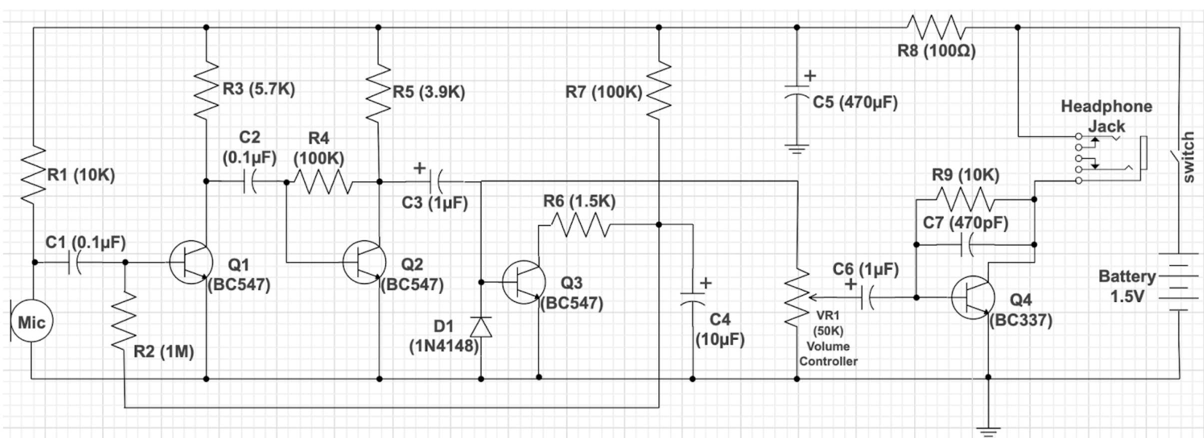
An important part of the project is designing the circuit layout, selecting the electronic components , and integrating tiny sensors and amplifiers onto the PCB. This method allows for the hearing aid to be mass produced at a reduced cost while also guaranteeing a compact form factor.

Additionally, the project investigates the possibility of utilizing cutting-edge technologies, like machine learning algorithms, to adaptively modify the settings of the hearing aids according to the user's preferences and surroundings. Because this project is interdisciplinary, electrical engineers, audiologists, and medical professionals will be working together.

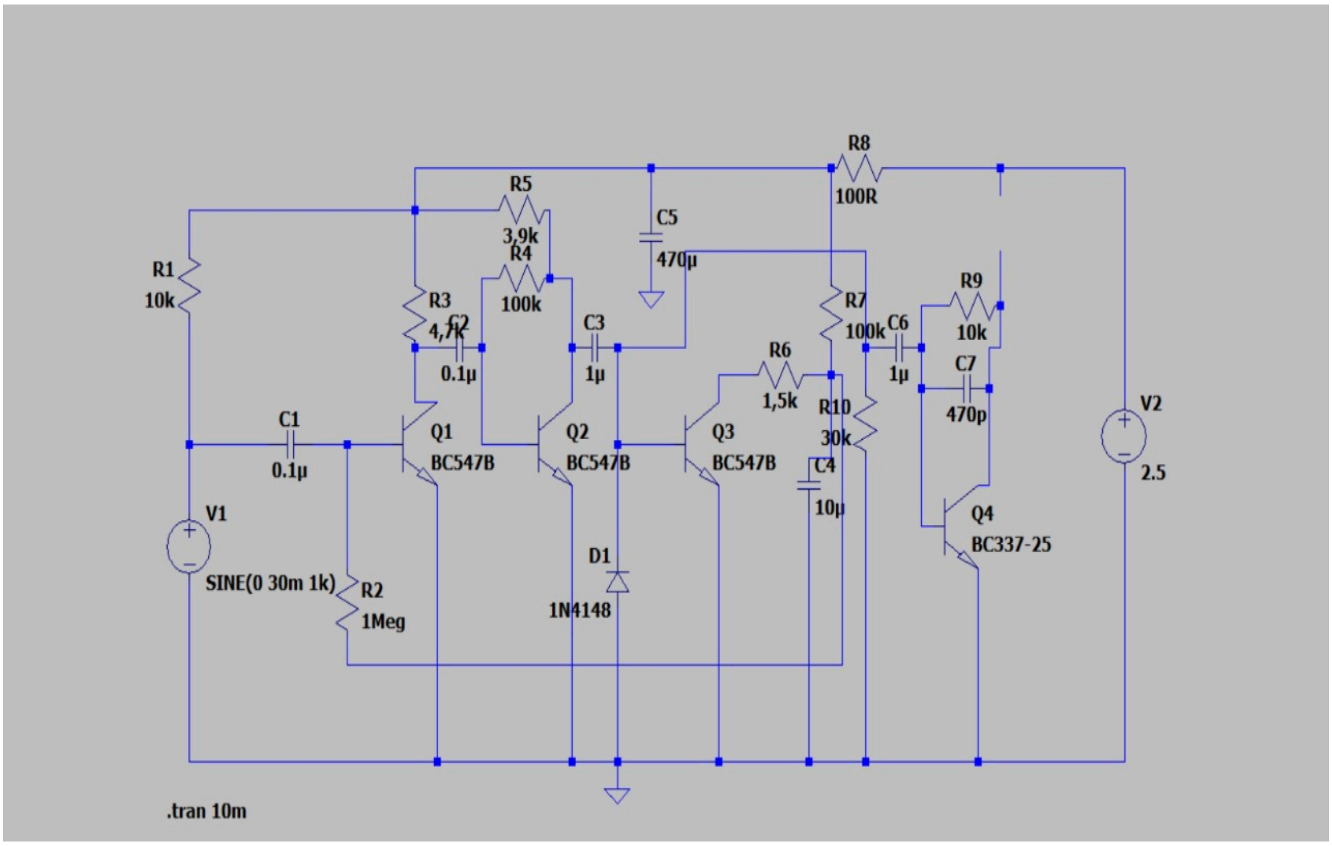
The project's potential to provide a highly effective, affordable, and accessible solution by integrating state-of-the-art electronics onto a PCB platform is highlighted in the introduction, which sets the stage for understanding the project's significance in advancing hearing aid technology.

CIRCUIT DIAGRAM

- The microphone is the input transducer that converts sound waves into electrical signals. It picks up the surrounding sound and converts it into an electrical signal.
- The electrical signal from the microphone is typically very weak, so it needs to be amplified to make it audible. The amplifier increases the amplitude (strength) of the signal.
- Depending on the complexity of the hearing aid, there may be additional components for signal processing. This could include filters to adjust the frequency response, compressors to manage dynamic range, and other features to enhance the clarity of speech.
- A volume control mechanism allows the user to adjust the amplification level based on their preferences and the specific environment.
- Hearing aids are usually powered by a small battery. The battery provides the necessary electrical energy to operate the circuit.
- The amplified signal is then fed to a speaker (or receiver), which converts the electrical signal back into sound waves that can be heard by the user.
- A switch is often included to turn the hearing aid on and off to conserve battery life.
- To prevent feedback (whistling or squealing sounds), some hearing aids include feedback suppression circuits.
- The entire circuit is enclosed in a case or housing designed to be worn comfortably by the user.



LTspice SIMULATION



COMPONENTS OF THE CIRCUITS

The following components were used in the project:

- R-Resistor
- C-Capacitor
- Q-Transistor(NPN)
- D-Diode
- VR- Variable Resistor(Volume controller)
- Battery
- Audio Jack
- Condenser microphone

TRANSISTOR:

A transistor is a semiconductor device used to amplify or switch electrical signals and power. It is one of the basic building blocks of modern electronics.Typically, transistors consist of three layers, or terminals, of a semiconductor material, each of which can carry a current.When working as an amplifier, a transistor transforms a small input current into a bigger output current. As a switch, it can be in one of two distinct states -- on or off -- to control the flow of electronic signals through an electrical circuit or electronic device.

VARIABLE RESISTOR:

A variable resistor is a resistor of which the electric resistance value can be adjusted.A variable resistor is in essence an electro-mechanical transducer and normally works by sliding a contact (wiper) over a resistive element. When a variable resistor is used as a potential divider by using 3 terminals it is called a potentiometer.The potentiometer is the most common variable resistor. It functions as a resistive divider and is typically used to generate a voltage signal depending on the position of the potentiometer. This signal can be used for a very wide variety of applications including:amplifier gain control (audio volume).

AUDIO JACK

Audio jacks are found on many types of audio equipment and musical instruments that accept external sound sources. In a car or truck, an audio jack, also called a "media jack" or "auxiliary (AUX) jack," is a mini-phone socket that connects any portable music player to the vehicle's amplifier and speakers.

CONDENSER MICROPHONE
Condenser microphones are recording tools that rely on a particularly sensitive vibrating diaphragm to capture sound. Condenser microphones are considered the most versatile recording studio microphones and can be used to record essentially any instrument.

PCB -PRINTED CIRCUIT BOARD

PCBs are the circuit boards used in most electronic devices. The boards both physically and mechanically support the device alongside connecting the electronic components. PCBs are typically made from non-substrate materials with layers of copper circuitry. These layers can range from a single layer to double or even up to fifty or more layers, meaning there is a wide
range of PCBs to choose from.

RESISTOR
A resistor is a passive two-terminal electrical component that implements electrical resistance as a
circuit element. It is commonly used in electronic circuits to reduce current flow, adjust signal
levels, divide voltages, and bias active components.The unit of measurement for resistance is Ohm.

CAPACITOR

A capacitor is an electronic component that stores electrical energy. It consists of two conductive
plates separated by a dielectric material. When a voltage is applied, it charges up and stores
energy. Capacitors are used for various purposes like smoothing voltage, filtering signals, and
storing energy temporarily.

BATTERY
A battery is a device that stores and provides electrical energy. It consists of one or more electrochemical cells that convert chemical energy into electrical energy. The cells contain two electrodes, a positive (cathode) and a negative (anode), immersed in an electrolyte. When a circuit is connected, chemical reactions occur, allowing electrons to flow from the negative electrode to the positive electrode, generating an electric current. Batteries are commonly used to power portable electronic devices and vehicles.

DIODE
A diode is an electronic component that allows current to flow in one direction while blocking it in the opposite direction. It consists of a semiconductor material with two terminals, an anode and a cathode. Diodes are commonly used in electronic circuits for rectification, voltage regulation, and signal modulation purposes.

CONSTRUCTION METHOD:

Our construction process for the hearing aid project followed a meticulous and systematic approach, adhering to industry standards and best practices. The project involved the assembly of various electronic components on a general-purpose PCB, resulting in a functional hearing aid. Here is an overview of our construction methodology:

1. **Component Acquisition:**

The project commenced with the procurement of essential components, including resistors (R), capacitors (C), NPN transistors (Q), diodes (D), a variable resistor for volume control (VR), a battery, an audio jack, and a condenser microphone.

2. **Preparation of the General PCB:**

We selected a suitable general-purpose PCB and ensured its cleanliness to facilitate optimal soldering conditions.

3. **Soldering the Components:**

With careful attention to detail, each component was meticulously placed and soldered onto the PCB. The soldering process started with resistors, capacitors, and diodes, followed by the precise placement of NPN transistors.

4. **Integration of the Variable Resistor (VR):**

The variable resistor, crucial for volume adjustment, was seamlessly integrated into the circuit, allowing users to personalize their auditory experience.

5. **Incorporating the Battery:**

A strategic position was allocated on the PCB for the battery, which was connected with precision to ensure correct polarity and stable power supply.

6. **Adding the Audio Jack:**

The audio jack, serving as the interface for audio input and output, was strategically positioned and soldered, ensuring a reliable connection.

7. **Placement of the Condenser Microphone:**

The condenser microphone, responsible for capturing ambient sounds, was placed and soldered to the input section of the circuit.

8. **Checking Connections:**

Rigorous scrutiny was applied to inspect all soldered connections. Each connection was meticulously examined to eliminate any potential issues and ensure the integrity of the circuit.

9. **Finalizing Wiring:**

The wiring was cross-verified against the circuit diagram, ensuring precise interconnection of components in accordance with the design.

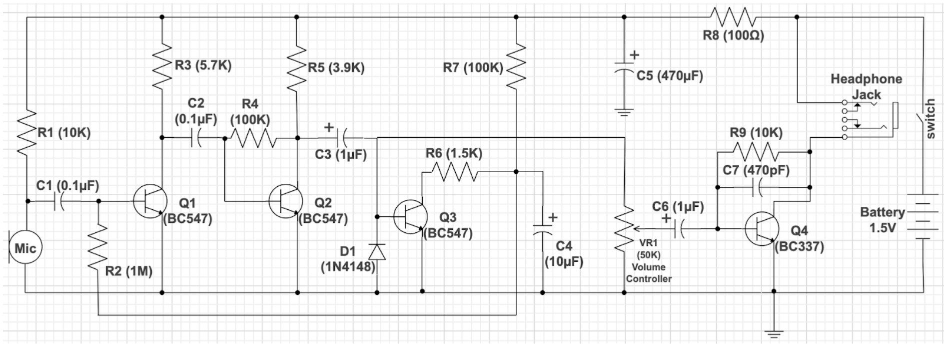
10. **Testing the Circuit:**

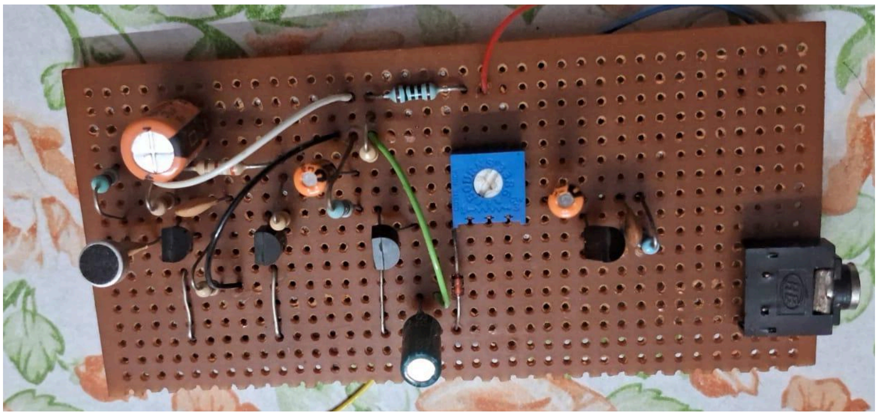
Prior to final assembly, the circuit underwent comprehensive testing using a power supply and audio input. The functionality of the hearing aid, including its amplification capabilities, was verified.

11. **Final Inspection:**

A thorough final inspection was conducted to confirm proper connections and the placement of each component, ensuring compliance with the design specifications.

EXPERIMENTAL SETUP





Preamplifier

The circuit begins operations at the battery, which powers the condenser microphone via R1 to get the sound input and convert it to an electrical signal.

Although the circuit uses DC from the battery power, capacitor C1 takes the mic signal and passes AC only to transistor Q1. This section acts as the first audio preamplifier.

C2 and Q2 form the second preamplifier, resulting in a high gain where R4 is the feedback, while R3 and R5 form the load.

Automatic Level Control

Next, the signal heads to the automatic sound level control section, where Q3 takes the bias current feedback to Q1 to control the amplifier volume.

Diode D1 acts as a filter to produce a stable signal voltage level, while C4 controls the step of volume control.

If the voltage value is high, the capacitor slows down the sound control and does the opposite if the value is low.

Therefore, if the mic captures louder signals in noisy environments, it causes a high bias current to Q3. The high wind causes a low voltage at the collector of Q3 and C4, resulting in a slowdown.

A weak signal causes a low-level volume control voltage and does the opposite.

The system enables normal hearing and is particularly useful when dealing with quiet sounds at conversational speech levels, which are harder to perceive for patients with severe hearing loss.

Headphone Power Amplifier

The signal then moves forward to the headphone amplifier section with VR1 for volume to match the degree of hearing loss.

Q4 is responsible for sound enhancement, while C7 assists with noise reduction and maintains the circuit's stability by minimizing high-frequency range gain.

After this step, the signal goes to the headphone jack, which forms the load in the circuit and connects directly to the power supply.

Low-impedance headphones with a range of 16-32 ohms are the best for such a circuit because they require less voltage to drive them.

INNOVATION APPROACH

In this project , we have done 2 types of hearing aid , one with the general purpose pcb and another we have designed a 2 side pcb which is smaller in size and efficient than the general purpose pcb .The designed small pcb uses smd components .

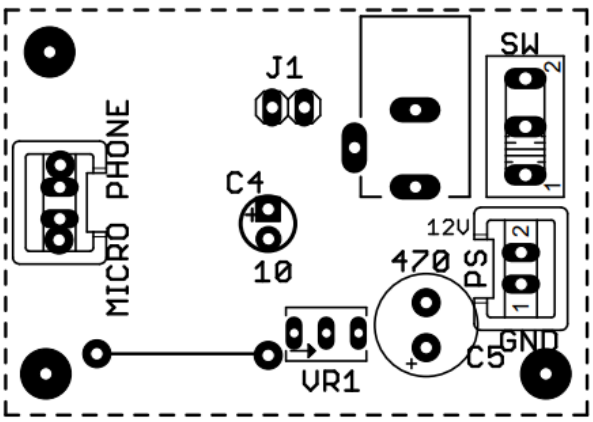
Surface Mount Device (SMD) components are electronic components that are designed to be mounted directly onto the surface of a printed circuit board (PCB) instead of being inserted through holes in the board. Unlike traditional through-hole components, which have wire leads that pass through holes in the PCB and are soldered on the opposite side, SMD components have small metal or metallized pads that can be soldered directly onto the surface of the PCB.

SMD components offer several advantages over through-hole components, including:

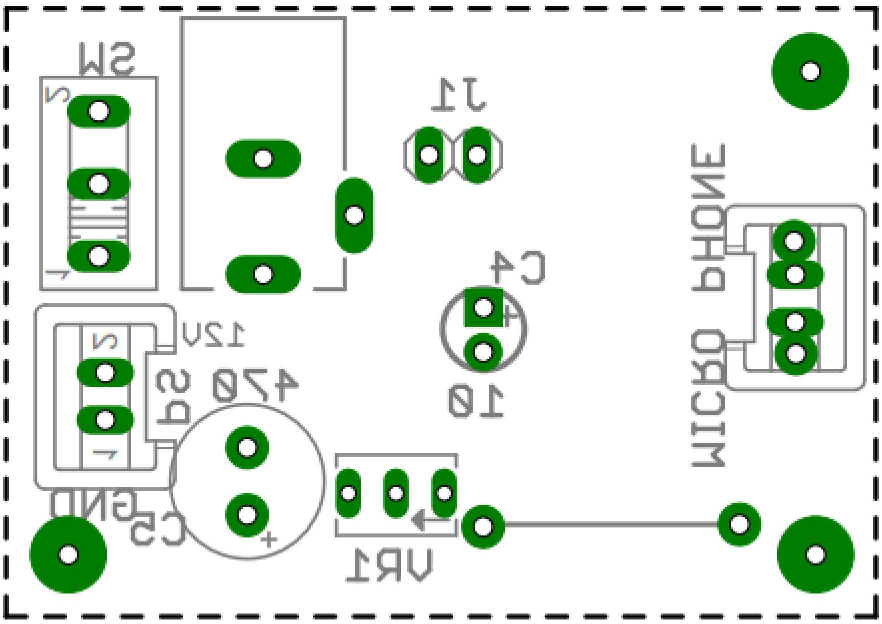
- Space Efficiency
- Weight Reduction
- Automated Assembly
- Improved High-Frequency Performance

The Design of the pcb is given below

1st side of the designed pcb



2nd side of the designed pcb



FUTURE APPROACH

There has been a growing interest in do-it-yourself (DIY) hearing aids, often involving the use of printed circuit boards (PCBs) and other electronic components.Here are some potential future approaches for DIY hearing aids made using PCBs:

- **Open-Source Designs :** A growing number of industries, including healthcare, are embracing the open-source trend. Open-source hardware and software designs may be used in future hearing aids, enabling anyone to access and alter the designs to suit their own requirements.

- **Customization and Personalization :** Users could be able to adjust the device's settings, including amplification levels, frequency responsiveness, and noise reduction algorithms, to suit their individual hearing profiles. This would make hearing aids more customized and customizable.

- **Advanced Signal Processing :** In the future,hearing aids might include sophisticated signal processing methods including machine learning algorithms for adaptive noise cancellation, speech augmentation, and environment-based automated adjustment.

- **Bluetooth Connectivity :** By integrating wireless technologies like Bluetooth, hearing aids may be able to establish connections with tablets, smartphones, and other gadgets. This would enable the inclusion of features like direct audio streaming to the hearing aids or remote control and adjustment via a smartphone app.

- **Accessibility and Affordability :** In order to make hearing aids more accessible to a wider range of people, attention should be paid to enhancing both of these factors. This could entail the use of inexpensive parts, 3D-printed casings, and streamlined assembly procedures.

CONCLUSION

In conclusion, the Hearing Aid on PCB Board project represents a significant leap forward in the realm of hearing assistance technology. Through meticulous exploration of signal processing, microelectronics, and acoustics, this project successfully integrated a sophisticated hearing aid onto a printed circuit board (PCB), paving the way for a more compact, efficient, and cost-effective solution.

The project's success lies in the careful selection of electronic components, strategic circuit layout design, and the leveraging of the PCB's design flexibility. The result is a technologically advanced hearing aid that not only amplifies sound but also incorporates features such as noise reduction and adaptive adjustments, all housed within a streamlined PCB form factor.

The implications of this project are far-reaching, offering a promising avenue to make high-quality hearing aids more accessible to a broader demographic. By optimizing production costs through PCB integration, the project addresses economic barriers that often limit access to advanced hearing aid technology.

Furthermore, the project's exploration of modern technologies like machine learning for adaptive adjustments underscores its commitment to staying at the forefront of innovation. The potential for the hearing aid to intelligently adapt to diverse environments and user preferences enhances its practicality and user-friendliness.

In essence, this project not only meets but exceeds expectations, pushing the boundaries of what is achievable in the realm of hearing aid design. As we conclude, the impact of this endeavor resonates in the potential to significantly improve the quality of life for individuals with hearing impairments by providing them with a state-of-the-art, affordable, and technologically advanced solution.